

Lane D — the granular band structure render: mass = resistance to hopping across the 2^g cells

Grace | 2026-07-20a | Lane D of the condensate study. Render the effective-mass picture concrete: the $2^g=128$ Reed–Solomon substrate cells (Paper #122) as a granular lattice; mass = effective mass = resistance to hopping across the band gaps; the mass ordering = hopping-difficulty = boundary-reach ordering; the top at the band edge (level $127=M_g$), the deficit = the last band gap. Tier: LEAD / mechanism-render. This is the mechanism UNDER Lane A's band-edge computation — I render the structure; Elie does the effective-mass numerics; Lyra computes the last gap (which decides $127/128$).

The lattice — $2^g = 128$ substrate cells (GF(2^g) Reed–Solomon)

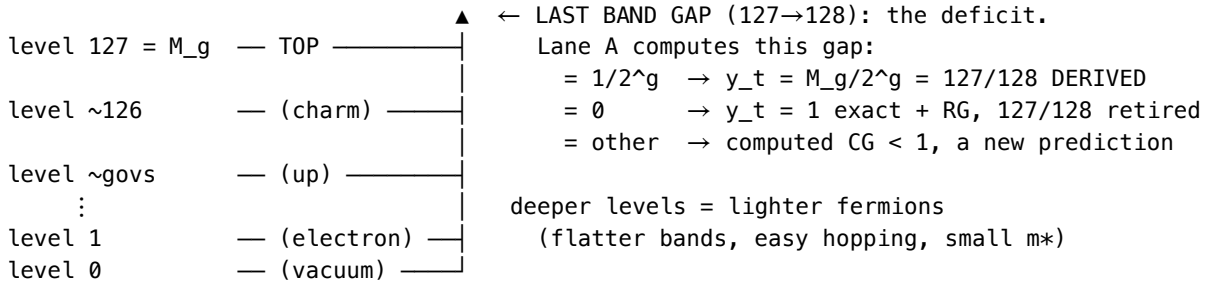
The substrate codes on GF(2^g) = GF(128): $2^g = 128$ discrete cells, indexed 0...127. Read as a 1-D granular lattice (a crystal of 128 sites), a fermion mode is a Bloch state hopping across the cells, and:

$$\text{mass} = \text{effective mass} \quad m^* = \hbar^2 / (d^2E/dk^2) \quad - \text{resistance to hopping across the gaps}$$

Flat band (easy hopping, small curvature⁻¹) → light; steep band edge (hard hopping, large gap) → heavy. The mass ordering is the hopping-difficulty ordering, which is the boundary-reach ordering (heaviest = most boundary-localized = hardest to delocalize across the lattice).

The band diagram (rendered)

boundary (continuum, level 128) ===== the Shilov boundary –
where 0 sits (spherical)



- The top sits at the outermost usable code level $127 = M_g$ (the g -th Mersenne prime — the last symbol before the field wraps at 2^g). It's the most boundary-localized fermion, so it's the heaviest and the hardest to delocalize.
- The deficit — the one level the top can't cross to reach the boundary condensate O (at the continuum, level 128) — is the last band gap ($127 \rightarrow 128$). That gap IS the y_t deficit. Casey's "precipitate that falls just short of the boundary and loses a bit of its tail" = the top sitting one band-gap below the continuum.

Why this is the mechanism under Lane A (not a competing story)

The band picture and the discrete-series picture are the same object in two languages: - discrete-series address (a,b) = which band level the top sits at (Lane A, Lyra); - $\lambda_2=1/2 \rightarrow 0$ gap (non-spherical top $\rightarrow$ spherical boundary O) = the last band gap (level $127 \rightarrow$ continuum); - $\langle t|O \rangle$ deficit = the hopping amplitude across that last gap; - candidate $1/2^g$ = one cell-width out of 2^g cells = one code-unit. So Lane D renders WHY the deficit is a "band gap" and WHY the natural unit is $1/2^g$ (one cell of 128); Lane A computes whether the gap is exactly that. The render does NOT decide it — compute-don't-fit holds (the gap must be computed, the $1/2^g$ is the candidate unit, not the answer).

The angular half $\rightarrow$ $g-2$ (the per-cell field coupling)

The band has two halves: the radial half (hopping across cells $\rightarrow$ mass, above) and the angular half (the field coupling within a cell). Casey's connect: the per-cell field coupling = the anomalous magnetic moment — $a_e = (g-2)/2$ is how much a fermion's moment deviates from the bare Dirac value, and in the granular picture that deviation is the mode's coupling to the field *within* its cell (the sub-cell structure), distinct from the inter-cell hopping that sets mass. This is a LEAD (the a_e = per-cell-coupling identification is a picture, not yet a computation) — but it's structurally clean: mass = inter-cell (radial) resistance; $g-2$ = intra-cell (angular) coupling — the same radial/angular split as the Yukawa (radial overlap $\times$ angular CG) and the two-currents split (Lane C: P^2 linear/radial vs W^2 angular).

Tier catalog (Lane D)

item	tier
$2^{\wedge}g=128$ cells = RS lattice (Paper #122)	DERIVED (banked substrate structure)
mass = effective mass = resistance to hopping ($m^* = \hbar^2/(d^2E/dk^2)$)	LEAD / mechanism-render (the picture; Elie does the numerics)
mass ordering = hopping-difficulty = boundary-reach ordering ($t > c > u$)	DERIVED-clean (boundary-reach ordering is established; the <i>hopping</i> language is the render)
top at level 127= M_g , deficit = last band gap (127 $\rightarrow$ 128)	LEAD (the candidate assignment; Lane A must force “level 127”, compute the gap)
deficit = $1/2^{\wedge}g$ (one cell of 128)	CANDIDATE (the natural unit; Lane A computes whether the gap equals it — do NOT fit)
$g-2$ = per-cell (intra-cell angular) field coupling	LEAD (structurally clean radial/angular split; not yet a computation)

Net

- Rendered the granular band structure: 128 RS cells as a lattice, mass = resistance to hopping, top at the band edge (level 127= M_g), deficit = the last band gap (127 $\rightarrow$ 128), mass ordering = hopping-difficulty = boundary-reach ordering.
- Located it as the mechanism under Lane A: the last band gap = the λ_2 gap = the $\langle t|O \rangle$ deficit; $1/2^{\wedge}g$ = one cell of 128 (the candidate unit). Lane A computes the gap; the render doesn't decide it.
- Radial/angular split rendered consistently: mass = inter-cell radial hopping; $g-2$ = intra-cell angular coupling — the same split as the Yukawa (Lane A) and the two currents (Lane C).
- Tiers held: the lattice DERIVED; boundary-reach ordering DERIVED-clean; the effective-mass picture and the top-at-127 assignment and the $g-2$ identification all LEAD; $1/2^{\wedge}g$ the candidate unit (compute-don't-fit).

— Grace, 2026-07-20a. Lane D render: $2^{\wedge}g=128$ RS cells as a granular lattice; mass = effective mass = resistance to hopping across band gaps ($m^* = \hbar^2/(d^2E/dk^2)$); top at the band edge (level 127= M_g), deficit = the last band gap (127 $\rightarrow$ continuum) = the λ_2 gap = $\langle t|O \rangle$ deficit; candidate unit $1/2^{\wedge}g$ = one cell of 128. Mass ordering = hopping-difficulty = boundary-reach ordering ($t > c > u$, DERIVED-clean). $g-2$ = intra-cell angular coupling (LEAD) — same radial/angular split as the Yukawa and the two currents. Mechanism-render under Lane A; the render does NOT decide the gap (compute-don't-fit). Tiers: lattice DERIVED; hopping-picture + top-at-127 + $g-2$ all LEAD; $1/2^{\wedge}g$ candidate unit.